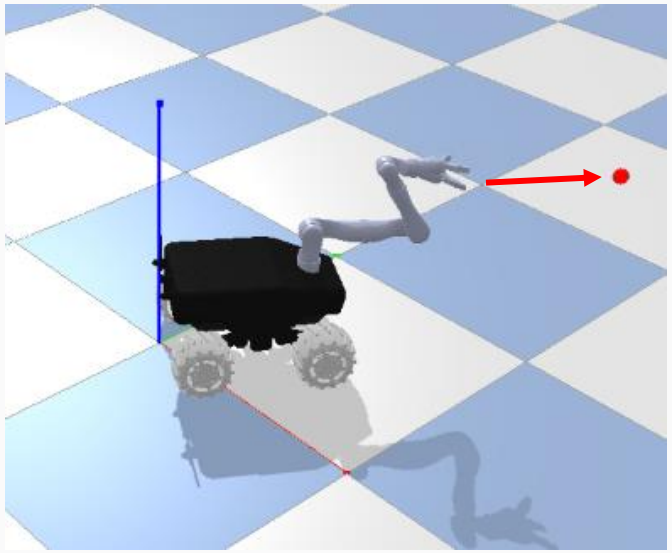
COMP0245 Assignment - 2026/27

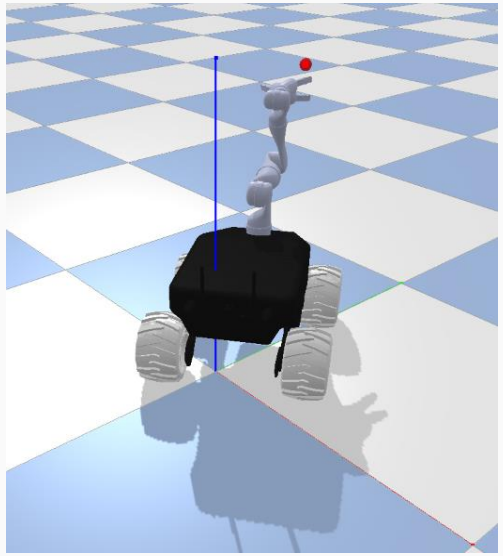
Francisco Vasconcelos
Valerio Modugno



Assignment Goal

- We provide a code template that moves the robot end-effector to a target cartesian position (x, y, z)
- We want you to train a machine learning model that replicates this behaviour and generalises to new targets and/or initial conditions (behaviour cloning)

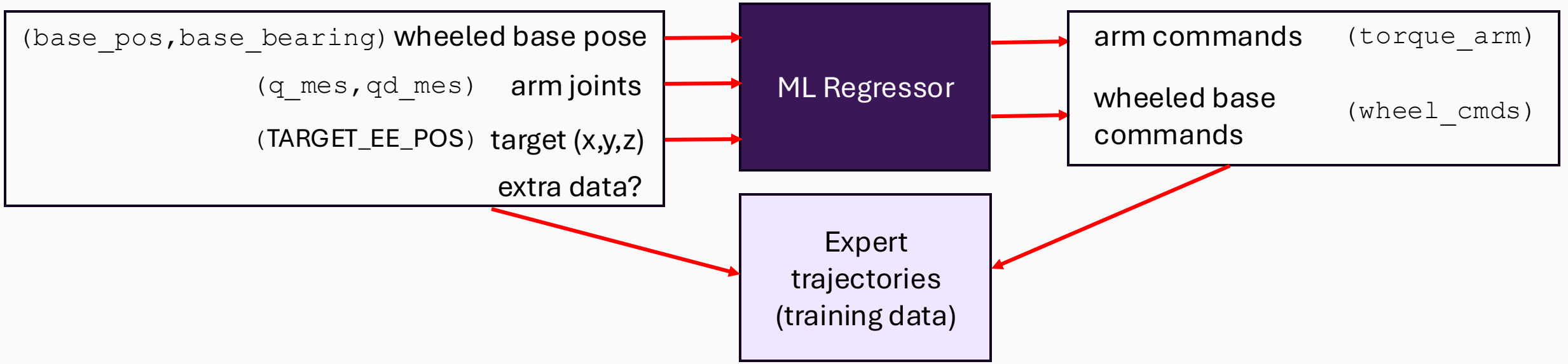




Outline of Assignment Task

- **Behaviour cloning - Fully supervised learning where "expert" trajectories act as training data**
- **You will run and record multiple simulations with different targets and/or conditions to generate "expert" trajectories**
- **You will train a supervised ML model that receives as input the robot state (base pose, arm joints) and target and generates as output commands to the robot.**

Outline of Assignment Task



- **Implement a regressor for this task**
- **Generate training data (expert trajectories)**
- **Design an evaluation methodology**
- **Justify your choices**

Methodology

- **Consider how to implement your solution:**
  - **Quantity and quality of data needed?**
  - **Would extra sensors/input data help?**
  - **Input data to regressor: single instant vs. temporal interval?**
  - **ML regressor architecture?**
  - **Training methodology and hyperparameters?**

Evaluation

- **Consider how to evaluate different factors:**
  - **How accurate is your solution?**
  - **How generalisable is your solution to different targets/initial conditions?**
  - **How robust is your solution to potential adversities?**
  - **How efficient is your solution (e. g. memory, parameters, computation time)?**

Timeline

- **20/10 (Tuesday):**
  - **Assignment Briefing.**
- **30/10-06/11:**
  - **Show us any preliminary or final solution running on your laptops (in-person during lab sessions).**
  - **Doesn't need to be final or fully working, just demonstrating that you are working towards the submission.**
  - **Not marked, but required for assessment completion.**
- **06/11 (Friday):**
  - **Report and code submission deadline**
  - **Marked**

Suggested milestones:

- **19/10-23/10 (week 1):**
  - **Initial training/testing data collected and validated**
- **26/10-30/10 (week 2):**
  - **ML model prototype completed, even if suboptimal**
  - **Report early draft**
- **02/11-06/11 (week 3):**
  - **ML model optimisation / additional experiments**
  - **Report completion**

Report Submission

- **Maximum 6 pages - use LaTeX IEEE template**
  - **Introduction**
    - **Briefly formulate your problem, state your assumptions and constraints**
  - **Methodology**
    - **Describe how you implemented your work**
    - **Justify choices you had to make**
  - **Results**
    - **Design appropriate experiments and comparisons**
    - **Describe the results you obtained (including tables, plots, etc)**
    - **Comment and interpret the results you obtained**
  - **Conclusion**
    - **Comment on limitations of method/experiments**
    - **How can method/experiments be improved**

Contribution statement Submission

- **Maximum 1 page**
  - **You can attach it as the 7th page of the report)**
  - **Summarise how each member of the group contributed to the submitted work**

Code submission

- **Include:**
  - **Readme file describing:**
    1. **How to run the code, and which outputs are generated**
    2. **Which external code sources were used and which parts of the code have been modified and/or coded from scratch.**
- **No need to include in Moodle submission:**
  - **Data**
  - **Pre-trained weights / checkpoints**

- **If you want to submit code that exceeds Moodle file size limit, you can include a github link to your code in the report.**

Marking Criteria

- **0-49%:**
  - **Largely incomplete work (report or code)**
  - **Fundamental methodology issues that invalidate the obtained results**
  - **Severe inconsistency between code and report methodology**

- **50-59%:**
  - **Major methodology issues that significantly affect the obtained results**
  - **Small degree of critical thinking (e. g. Code resources are just used as “black-box”, without demonstrable understanding of how they are fully utilised and/or adapted to the problem at hand)**
  - **Lack of justification/reasoning for the majority of method design choices and experiments**
  - **Inconsistency in important elements of code and report methodology**

Marking Criteria

- **60-69%**
  - **Methodology is overall correct, but with some technical issues that affect the obtained results to a limited degree**
  - **Important method components are not fully reproducible, either due to unclear report or code documentation/organisation.**
  - **Justifications/discussions with significant amount of generic statements that do not reflect specific data/model/results of the submitted work.**
- **70-79%**
  - **Only extremely minor technical issues present, with very subtle effects on the obtained results. The code is very well documented and the work is reproducible. The work demonstrates excellent understanding of the implemented methodology, and how code resources are used/modified.**
- **80-100%**
  - **On top of the 70-80% criteria, the work shows well utilized/justified novel ideas in the method design or the experimental validation**

Marking Criteria

- **By default, we assume that group members contribute a similar amount work towards the final submission, and thus all members have the same mark (based on a single report+code submission)**

- **If a significant imbalance in contribution is affecting the work in your group, please let us know before the submission deadline. Adjustments to individual marks may be applied in such cases.**